

CustomerIQ

Product Requirements Document & Official Project Record

Version 2.0 | Project status: End of Day 3 | Data foundation, purchase behavior, and churn methodology completed

Purpose of this document: Preserve the agreed scope, methodology, completed work, actual observed results, limitations, risks, and next steps so the CustomerIQ project can be continued without losing analytical context.

1. Executive Summary

CustomerIQ is a customer intelligence and retention analytics project built from transaction-level retail data. The intended business chain is: **Customer Behavior** → **Insight** → **Risk** → **Priority** → **Action** → **Monitoring**. The project is business-first rather than model-first.

By the end of Day 3, the project has progressed from raw transaction data to a validated customer-level analysis dataset with a time-separated observed return/no-return outcome. The source data was cleaned, customer purchase behavior was studied, a censoring-aware temporal split was chosen, pre-cutoff customer features were created, and a 90-day look-forward outcome was generated without using future transactions in feature construction.

2. Product Vision and Problem

Working description: AI-Powered Customer Intelligence & Retention Analytics Engine.

Transaction data records what customers bought, but does not directly answer which customers are valuable, which have changed behavior, who should be prioritized, why a customer appears at risk, or what action should be considered. CustomerIQ is intended to transform transaction history into interpretable customer intelligence and, later, predictive risk, prioritization, recommendations, and an executive-facing dashboard.

Area	Definition
Primary users	Marketing analysts, CRM/retention teams, customer analytics teams, e-commerce/retail managers
Secondary users	Account managers, business analysts, executives
Core jobs to be done	Understand purchase behavior; identify disengagement; prioritize limited retention resources; explain risk
Demonstration context	UK online retail data with meaningful wholesale/B2B characteristics; not presented as a fictional beauty c

3. Dataset Evidence Base

Primary dataset: UCI Online Retail II, used as the demonstration dataset.

Raw dataset metric	Observed result
Rows / columns	1,067,371 rows / 8 columns
Date range	2009-12-01 07:45:00 to 2011-12-09 12:50:00
Unique customers before cleaning	5,942

Raw dataset metric	Observed result
Unique invoices before cleaning	53,628
Columns	Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country
Missing Description	4,382
Missing Customer ID	243,007 (22.77%)
Exact duplicate rows	34,335

Important interpretation rule: because the dataset contains wholesale/B2B-style purchasing behavior, Average Order Value, repeat rates, and purchase rhythms should not be generalized as typical consumer e-commerce benchmarks.

4. Day 1 — Data Foundation and Cleaning

- Loaded the raw dataset in Python and validated shape, columns, data types, missing values, duplicates, date range, customer count, and invoice count.
- Investigated duplicate records before removing exact duplicates.
- Created a cleaned working dataset by removing exact duplicates, rows without Customer ID, cancelled invoices beginning with C, non-positive quantities, and non-positive prices.
- Converted InvoiceDate to datetime for temporal analysis.
- Created Revenue = Quantity × Price as a derived analytical feature.
- Saved the cleaned reusable dataset as the processed source for later notebooks.

Day 1 business metric	Actual result
Cleaned dataset	779,425 rows and 8 source fields
Total Revenue	£17,374,804.27
Total Customers	5,878
Total Orders	36,969
Average Order Value	£469.98
Repeat Customers	4,255
One-Time Customers	1,623
Repeat Customer Rate	72.39%

Day 1 caveat: The AOV and repeat rate are valid calculations for this cleaned dataset but should not be framed as typical consumer-retention benchmarks because of the dataset mix.

5. Day 2 — Purchase Behavior and Temporal Methodology

- Built the customer-order distribution using unique invoices per customer.
- Measured minimum, median, average, and maximum order frequency.
- Examined inter-purchase intervals rather than forcing an arbitrary churn threshold.
- Segmented customers by provisional order-frequency tiers and explicitly treated tier boundaries as a starting analytical simplification rather than a discovered universal truth.

- Addressed right-censoring by separating historical observation from a fixed future look-forward period.
- Selected 2011-09-10 as the provisional observation cutoff, leaving a 90-day validation period through the dataset end on 2011-12-09.

Day 2 result	Actual result
Minimum orders per customer	1
Median orders	3.0
Average orders	6.29
Maximum orders	398
Observed inter-purchase intervals	31,091
Median interval	24.74 days
90th percentile interval	134.88 days
One-time tier	1,623 customers
2–3 orders	1,608 customers
4–9 orders	1,673 customers
10+ orders	974 customers
Observation cutoff	2011-09-10
Look-forward period	90 days through 2011-12-09

6. Temporal Methodology Decision

Observation period: transactions strictly before 2011-09-10. **Validation period:** 2011-09-10 through 2011-12-09.

The 90-day window is an **observation/validation window**. At this stage it is not claimed to be a universally correct business definition of churn. The dataset has a hard end date, so customers near the end have limited opportunity for later observed purchases. This right-censoring risk is explicitly documented.

Customers contribute different numbers of purchase intervals; customers with few orders provide noisier evidence about their natural rhythm. Wholesale/B2B purchasing patterns further limit direct generalization to consumer e-commerce.

7. Day 3 — Customer-Level Churn Methodology

Day 3 converted transaction-level history into one customer-level analysis record for the eligible repeat-purchase cohort. Eligibility was based on at least 2 unique invoices before the cutoff. Customers with only one pre-cutoff order were counted and documented as an insufficient-history / one-time group rather than silently deleted.

Population	Actual result
Total customers before cutoff	5,281
One-time pre-cutoff customers	1,576 (29.84%)
Eligible customers with 2+ pre-cutoff orders	3,705 (70.16%)
Returned during validation	1,973
No observed return during validation	1,732

Population	Actual result
Return rate	53.25%
Provisional observed churn rate	46.75%

8. Customer-Level Features Created

Using **only pre-cutoff transactions**, one row was created for each of the 3,705 eligible customers.

Feature	Definition
Customer ID	Unique customer identifier
Frequency	Number of unique invoices before cutoff
Monetary	Total Revenue before cutoff
First Purchase Date	Earliest purchase date before cutoff
Last Purchase Date	Latest purchase date before cutoff
Customer Lifespan	Days between first and last purchase
Recency	Days between last purchase and cutoff date
Average Purchase Interval	Average time gap between consecutive purchases/orders
Median Purchase Interval	Median time gap between consecutive purchases/orders
returned_in_validation	1 if at least one purchase occurred in validation; otherwise 0
observed_churn	1 if no observed purchase occurred during the 90-day validation period; otherwise 0

9. Day 3 Customer Behavior Summary

Metric	Actual result
Average Frequency	7.76 unique orders
Median Frequency	4 unique orders
Average Monetary Value	£3,615.15
Median Monetary Value	£1,333.70
Average Customer Lifespan	319.5 days
Average Recency	155.5 days before cutoff

10. Data Leakage Audit

A core Day 3 requirement was preventing future information from entering the historical customer features.

- All customer features were built using transactions before 2011-09-10.
- The return/no-return outcome was built only from transactions during 2011-09-10 through 2011-12-09.
- No future purchase was used to construct Frequency, Monetary, Recency, Lifespan, or purchase-interval features.

- Revenue was calculated from Quantity × Price inside the relevant temporal window.
- Final validation confirmed one row per eligible customer.

Result: Leakage Check — PASS. All predictive features were constructed using only pre-cutoff transaction history.

11. Final Day 3 Dataset and Validation

Validation item	Actual result
Output file	Data/Processed/customer_churn_analysis.csv
Dataset shape	3,705 rows × 11 columns
Unique Customer IDs	3,705
Duplicate Customer IDs	0
Missing values across listed features	0
Notebook execution	PASS — all code cells executed successfully
Output CSV	Successfully saved and verified

The processed source dataset was not overwritten. Revenue was recreated in memory where required. The final customer-level dataset is a separate output artifact.

12. Current Project Architecture

Current analytical flow:

Raw Online Retail II CSV → Day 1 cleaning and validation → online_retail_cleaned.csv → Day 2 purchase behavior and temporal methodology → Day 3 customer feature engineering and validation outcome → customer_churn_analysis.csv → next analytical stages.

Artifact	Role
01 / Day 1 notebook	Data validation, cleaning, revenue and initial business metrics
02_purchase_behavior.ipynb	Customer order distribution, purchase intervals, frequency tiers, temporal cutoff
03_churn_methodology.ipynb	Eligible cohort, pre-cutoff features, validation outcome, leakage checks
online_retail_cleaned.csv	Cleaned transaction-level source for downstream analysis
customer_churn_analysis.csv	Customer-level Day 3 analysis dataset

13. Known Risks and Assumptions

- **Right-censoring:** the dataset ends on 2011-12-09; the temporal split mitigates but does not eliminate the limitation.
- **Provisional churn outcome:** no return in 90 days is an observed outcome for this study, not a universal definition of permanent churn.
- **Wholesale/B2B behavior:** customer purchase cadence may differ materially from consumer e-commerce.
- **Low-order uncertainty:** customers with fewer purchases provide less evidence about typical purchase intervals.

- **Recommendation causality:** future recommendations must not claim they are proven to reduce churn without treatment/outcome evidence.
- **Scope control:** additional complexity such as deep learning, XGBoost, survival analysis, or K-Means is not required unless later evidence justifies it.

14. What Is Complete vs Remaining

Status	Work item
Completed	Data selection, validation, cleaning, processed transaction dataset
Completed	Revenue calculation and baseline business KPIs
Completed	Repeat-customer analysis
Completed	Customer order distribution and purchase-interval analysis
Completed	Frequency-tier analysis and censoring-aware temporal cutoff
Completed	Customer-level pre-cutoff feature engineering
Completed	90-day return/no-return outcome and leakage audit
Next	Compare returned vs no-return customers to identify behavioral signals
Planned	RFM and interpretable customer segmentation
Planned	Churn modeling and model evaluation
Planned	Value × risk prioritization
Planned	Recommendation/decision framework and optional GenAI explanation layer
Planned	Power BI dashboard, README, case study, screenshots, demo, interview preparation

15. Day 4 Objective

The immediate next analytical objective is to understand what separates customers who returned from customers who did not return during the validation period. The first comparison should focus on Frequency, Monetary value, Recency, Customer Lifespan, Average Purchase Interval, and Median Purchase Interval.

The business question is: **Which observable behaviors before the cutoff are associated with a customer later returning versus showing no observed return?**

Machine learning should not be the immediate next step until these basic behavioral patterns are understood and checked for plausibility.

16. Product Success Criteria

- Reproducible data pipeline with documented cleaning decisions.
- Defensible time-aware methodology and explicit leakage control.
- Interpretable customer segmentation.
- Leakage-aware churn/return modeling evaluated beyond simple accuracy.
- Customer value combined with risk for practical prioritization.
- Traceable, explainable recommendation logic.

- Focused Power BI dashboard answering business questions.
- Clear GitHub README and portfolio narrative.
- Ability to explain major methodological choices and limitations in an interview.

17. Current Progress Assessment

The original PRD recorded the project at approximately 12–15% completion at the end of the earlier foundation stage. Since then, Day 2 and Day 3 have added the purchase-behavior methodology, explicit temporal split, customer-level features, observed outcome construction, and leakage validation. The project has materially progressed beyond the original foundation-only state, but major intelligence, segmentation, modeling, prioritization, recommendation, and dashboard layers remain.

A precise percentage should not be treated as an analytical metric. The defensible status is: **Days 1–3 are complete; the data and methodology foundation is strong; the remaining work is the insight, segmentation, predictive, prioritization, action, and presentation layers.**

18. Interview Narrative

I built CustomerIQ, a customer intelligence and retention analytics system using transaction-level online retail data. I began by validating and cleaning more than one million raw records, documented missing identifiers and duplicates, and created a reusable cleaned transaction dataset. I then analyzed customer order frequency and purchase intervals to avoid choosing a churn rule blindly. To address the dataset end-date problem, I created a fixed observation cutoff and 90-day look-forward window. Finally, I transformed pre-cutoff transaction history into a customer-level dataset for 3,705 eligible repeat customers and created a provisional observed return/no-return outcome while explicitly checking for data leakage. The next stage is to identify the behavioral differences between returning and non-returning customers before moving into segmentation and predictive modeling.

19. Official Change Log

Version	Project state	Change
1.0	End of Day 1 / before Day 2	Initial PRD documenting product scope, raw and cleaned data, business metrics
2.0	End of Day 3	Added completed Day 2 purchase behavior and temporal methodology; added c

End of official project record — CustomerIQ PRD v2.0